

UNIVERSIDAD DEL VALLE DE GUATEMALA

Facultad de Ingenieria - Cifrado de Informacion

PROYECTO 1

Cifrados de Flujo

Desafíos en Seguridad

CTF One Piece Symmetric Cipher

Reporte Tecnico de Resolución

Curso: Cifrado de información - Universidad del Valle de Guatemala

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Repositorio: https://github.com/gabrielpaz2003/Proyecto1_Cifrados.git

1. Introduccion

En este proyecto se resolvieron cuatro retos practicos de cifrados de flujo inspirados en One Piece. Cada reto consistio en localizar archivos dentro de una estructura de carpetas generada aleatoriamente, extraer datos cifrados, aplicar el algoritmo correspondiente y documentar la flag y el texto del poneyglyph recuperado.

Los algoritmos abarcados van desde el clasico XOR hasta el moderno ChaCha20, pasando por RC4 y un cifrado personalizado basado en PRNG, permitiendo comparar sus fortalezas y debilidades.

2. Entorno de trabajo

Sistema operativo / terminal: Ubuntu WSL (Windows Subsystem for Linux)

Herramientas principales: find, 7z, exiftool, xxd, perl, python3, Docker Compose

Configuración inicial

```
pip install -r resources/requirements.txt
```

```
python generate_challenges.py
```

Captura 1 - Generación de retos y contenedores Docker activos.

```
PS C:\Gabriel\UVG\9 SEMESTRE\CIFRADO DE INFORMACIÓN\Proyecto1\ctf_onepice_symmetric_cipher> pip install -r resources/requirements.txt
Requirement already satisfied: pillow in C:\Users\gabriel\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from -r resources/requirements.txt (line 1)) (12.2.0)
Requirement already satisfied: numpy in C:\Users\gabriel\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from -r resources/requirements.txt (line 2)) (2.4.4)
Requirement already satisfied: piexif in C:\Users\gabriel\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from -r resources/requirements.txt (line 3)) (1.1.3)
Requirement already satisfied: pyzipper in C:\Users\gabriel\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from -r resources/requirements.txt (line 4)) (0.3.6)
Requirement already satisfied: pycryptodome in C:\Users\gabriel\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from -r resources/requirements.txt (line 5)) (3.23.0)
Requirement already satisfied: pycryptodomex in C:\Users\gabriel\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pyzipper->-r resources/requirements.txt (line 4)) (3.23.0)

[notice] A new release of pip is available: 26.0.1 -> 26.1.1
[notice] To update, run: C:\Users\gabriel\AppData\Local\Python\pythoncore-3.14-64\python.exe -m pip install --upgrade pip
PS C:\Gabriel\UVG\9 SEMESTRE\CIFRADO DE INFORMACIÓN\Proyecto1\ctf_onepice_symmetric_cipher> python generate_challenges.py
Ingrese su carné: 221887
Laberinto generado con éxito.
Laberinto generado con éxito.
Laberinto generado con éxito.
Laberinto generado con éxito.
Retos generados con éxito!
PS C:\Gabriel\UVG\9 SEMESTRE\CIFRADO DE INFORMACIÓN\Proyecto1\ctf_onepice_symmetric_cipher>
```

```
docker compose up -d --build
```

```
docker ps
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

#41 DONE 0.2s

#42 [luffy_image] resolving provenance for metadata file
#42 DONE 0.0s

#43 [nami_image] resolving provenance for metadata file
#43 DONE 0.1s

#44 [zoro_image] resolving provenance for metadata file
#44 DONE 0.2s
[+] up 9/9
✓ Image ctf_onepice_symmetric_cipher-nami_image Built
✓ Image ctf_onepice_symmetric_cipher-usopp_image Built
✓ Image ctf_onepice_symmetric_cipher-luffy_image Built
✓ Image ctf_onepice_symmetric_cipher-zoro_image Built
✓ Network ctf_onepice_symmetric_cipher_ctf_network Created
✓ Container usopp_challenge Created
✓ Container zoro_challenge Created
✓ Container luffy_challenge Created
✓ Container nami_challenge Created
PS C:\Gabriel\UVG\9 SEMESTRE\CIFRADO DE INFORMACIÓN\Proyecto1\ctf_onepice_symmetric_cipher>
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
4abcab745168	ctf_onepiece_symmetric_cipher-usopp_image	"tail -f /dev/null"	47 seconds ago	Up 45 seconds	0.0.0.0:2204->22/tcp, [::]:2204->22/tcp, 0.0.0.0:8084->8080/tcp, [::]:8084->8080/tcp	usopp_challenge
4e3d2c36bd44	ctf_onepiece_symmetric_cipher-luffy_image	"tail -f /dev/null"	47 seconds ago	Up 45 seconds	0.0.0.0:2201->22/tcp, [::]:2201->22/tcp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp	luffy_challenge
38cd763c448	ctf_onepiece_symmetric_cipher-nami_image	"tail -f /dev/null"	47 seconds ago	Up 45 seconds	0.0.0.0:2203->22/tcp, [::]:2203->22/tcp, 0.0.0.0:8083->8080/tcp, [::]:8083->8080/tcp	nami_challenge
daa493f66a71	ctf_onepiece_symmetric_cipher-zoro_image	"tail -f /dev/null"	47 seconds ago	Up 45 seconds	0.0.0.0:2202->22/tcp, [::]:2202->22/tcp, 0.0.0.0:8082->8080/tcp, [::]:8082->8080/tcp	zoro_challenge

3. Metodología general

Para cada reto se siguió el mismo flujo de trabajo:

- 1 Buscar archivos flag.txt cuyo contenido sea hexadecimal puro (descartar señuelos con texto legible).
- 2 Localizar el ZIP correcto data_*.zip extrayendo su imagen y revisando el campo EXIF Artist.
- 3 Si Artist contiene hex, descifrar con XOR usando el carne del estudiante para obtener el poneglyph.
- 4 Aplicar el algoritmo del reto al ciphertext para recuperar la flag, que a su vez desbloquea el siguiente ZIP.

Reto 1 - Monkey D. Luffy

XOR

Objetivo: recuperar la flag cifrada con XOR (clave = carne). Contraseña inicial del ZIP: onepiece

Comandos principales

```
docker exec -it luffy_challenge bash
```

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

luffy@4e3d2c36bd44:~$ ls -la
total 24
drwxr-x--- 1 luffy luffy 4096 May 24 04:05 .
drwxr-xr-x 1 root  root  4096 May 24 04:05 ..
-rw-r--r-- 1 luffy luffy  220 Jan  6  2022 .bash_logout
-rw-r--r-- 1 luffy luffy 3771 Jan  6  2022 .bashrc
-rw-r--r-- 1 luffy luffy  807 Jan  6  2022 .profile
drwxr-xr-x 1 luffy luffy 4096 May 24 03:51 ONEPIECE
luffy@4e3d2c36bd44:~$ ls ONEPIECE/
Dressrosa  East Blue  Skypiea  Whole_Cake_Island  Zou
luffy@4e3d2c36bd44:~$

```

Después de realizar varias búsquedas dentro de los archivos encontramos la flag

```
cd ONEPIECE
```

```
ls
```

```
cd Whole_Cake_Island
```

```
ls
```

```
cd Whole_Cake_Chateau
```

```
ls
```

```
cd Casa_de_Pudding
```

```
ls -la
```

```
cat flag.txt
```

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

luffy@4e3d2c36bd44:~$ cd ONEPIECE
luffy@4e3d2c36bd44:~/ONEPIECE$ ls
Dressrosa  East_Blue  Skypiea  Whole_Cake_Island  Zou
luffy@4e3d2c36bd44:~/ONEPIECE$ cd Whole_Cake_Island
luffy@4e3d2c36bd44:~/ONEPIECE/Whole_Cake_Island$ ls
Cacao_Island  Caramel_Mountain  Liqueur_Island  Sweet_City  Whole_Cake_Chateau
luffy@4e3d2c36bd44:~/ONEPIECE/Whole_Cake_Island$ cd Whole_Cake_Chateau
luffy@4e3d2c36bd44:~/ONEPIECE/Whole_Cake_Island/Whole_Cake_Chateau$ ls
Casa_de_Big_Mom  Casa_de_Katakuri  Casa_de_Pudding
luffy@4e3d2c36bd44:~/ONEPIECE/Whole_Cake_Island/Whole_Cake_Chateau$ cd Casa_de_Pudding
luffy@4e3d2c36bd44:~/ONEPIECE/Whole_Cake_Island/Whole_Cake_Chateau/Casa_de_Pudding$ ls -la
total 16
drwxr-xr-x 1 luffy luffy 4096 May 24 03:51 .
drwxr-xr-x 1 luffy luffy 4096 May 24 03:51 ..
-rwxr-xr-x 1 luffy luffy 151 May 24 03:51 .marker_238
-rwxr-xr-x 1 luffy luffy 74 May 24 03:51 flag.txt
luffy@4e3d2c36bd44:~/ONEPIECE/Whole_Cake_Island/Whole_Cake_Chateau/Casa_de_Pudding$ cat flag.txt
747e70776705510b50540e00535405510106040106030f05070006065b56040005045a0705luffy@4e3d2c36bd44:~/ONEPIECE/Whole_Cake_Island/Whole_Cake_Chateau/Casa_de_Pudding$

```

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

luffy@4e3d2c36bd44:~/ONEPIECE/East_Blue/Romance_Dawn/Casa_de_Luffy$ python3 -c "
hex_string = "71405e357535b5e4104c5640554445d5312469955187648533514b4353127a595650565d5c105a52515344435d175d5411594c4412625e5e5d505e4b41581417455a58535017515d5f44595e5c5755105159545d435d59435b5d5f1057591246995518405
a5743559555d47454318585412615c4d435d5c1d10"
key = "221807"
data = bytes.fromhex(hex_string)
key_bytes = key.encode()
result = bytes([data[i] ^ key_bytes[i % len(key_bytes)] for i in range(len(data))])
print(result.decode())
"
Crocodile targeted the Arabasta Kingdom because of its Poneglyph, which contained information on the whereabouts of Pluton,
luffy@4e3d2c36bd44:~/ONEPIECE/East_Blue/Romance_Dawn/Casa_de_Luffy$

```

Resultados

Flag encontrada: **FLAG_2c9ad67af4a916373725276ca6244b07**

Poneglyph extraido:

"Crocodile targeted the Arabasta Kingdom because of its Poneglyph, which contained information on the whereabouts of Pluton,"

Análisis técnico — Reto Luffy (XOR)

Se exploró el laberinto de carpetas del contenedor navegando con ls y cd hasta encontrar el flag.txt real, identificándolo porque su contenido era hexadecimal puro, a diferencia de los archivos falsos que contenían texto legible. Se descifró aplicando XOR con el carné como clave usando Python. Para el poneyglyph, se extrajo el ZIP con contraseña onepiece, se leyeron los metadatos EXIF de la imagen obteniendo otro valor hexadecimal cifrado, y se descifró con el mismo método XOR.

Reto 2 - Roronoa Zoro

RC4

Objetivo: descifrar la flag con RC4. Clave = carne. Contraseña del ZIP: flag de Luffy.

Comandos principales

```
cd ~/ONEPIECE/Skypiea/Shandora/Casa_de_Calgara
```

```
ls -la
```

```
cat flag.txt
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

zorc@daa493fb6a71:~$ cd ~/ONEPIECE/Skypiea/Shandora/Casa_de_Calgara
zorc@daa493fb6a71:~/ONEPIECE/Skypiea/Shandora/Casa_de_Calgara$ ls -la
total 16
drwxr-xr-x 1 zoro zoro 4096 May 24 03:51 .
drwxr-xr-x 1 zoro zoro 4096 May 24 03:51 ..
-rwxr-xr-x 1 zoro zoro 129 May 24 03:51 .marker_238
-rwxr-xr-x 1 zoro zoro 74 May 24 03:51 flag.txt
zorc@daa493fb6a71:~/ONEPIECE/Skypiea/Shandora/Casa_de_Calgara$ cat flag.txt
39065e9fc94ea6d95dbf56b51d500236e0eb705136d2c1c86cb27e81dc7bc3372ea7c9738ezorc@daa493fb6a71:~/ONEPIECE/Skypiea/Shandora/Casa_de_Calgara$
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

Type = zip
Physical Size = 52475

Everything is Ok

Size: 52482
Compressed: 52475
Artist : 535c55185b45575345555c17465a54107a56405d40455d17655d435b4b17414b5f54515453465410515912535f10594346575c404c17465d11524a5e5c55115457405c1245585d177340585259444653117b515955565e5d18565c561
15354565b5f11605442465d5f1c18525f425d5f415e5c55116257555b5c114457174057505418435a571160575957555d49485f
zorc@daa493fb6a71:~/ONEPIECE/East_Blue/Syrup_Village/Casa_de_Merry$ python3 -c "
hex_string = '535c55185b45575345555c17465a54107a56405d40455d17655d435b4b17414b5f54515453465410515912535f10594346575c404c17465d11524a5e5c55115457405c1245585d177340585259444653117b515955565e5d18565c56115354565b5f11605442465d5f1c18525f425d5f415e5c55116257555b5c114457174057505418435a571160575957555d49485f'
key = '221007'
data = bytes.fromhex(hex_string)
key_bytes = key.encode()
result = bytes([data[i] ^ key_bytes[i % len(key_bytes)] for i in range(len(data))])
print(result.decode())
"
and created the Baroque Works syndicate in an attempt to bring down the Arabasta Kingdom and claim Pluton, employing Robin to read the Poneyglyph
zorc@daa493fb6a71:~/ONEPIECE/East_Blue/Syrup_Village/Casa_de_Merry$
```

Resultados

Flag encontrada: **FLAG_aeb36ce81cda75a776a0088a4c8a35f2**

Poneglyph extraído: ***"and created the Baroque Works syndicate in an attempt to bring down the Arabasta Kingdom and claim Pluton, employing Robin to read the Poneglyph"***

Análisis técnico — Zoro

Se navegó el laberinto de carpetas del contenedor con ls y cd hasta encontrar el flag.txt real, identificándolo por su contenido hexadecimal puro. Se descifró usando RC4, que es un cifrado de flujo que genera una secuencia de bytes pseudoaleatoria a partir de la clave (el carné) y la combina con el texto cifrado mediante XOR. Para el poneglyph, se extrajo el ZIP usando la flag de Luffy como contraseña, se leyó el campo Artist de los metadatos EXIF de la imagen y se descifró con XOR usando el carné.

Reto 3 - Usopp

PRNG
Custom

Objetivo: descifrar la flag con el PRNG de Python (semilla fija 1234). Contraseña del ZIP: flag de Zoro.

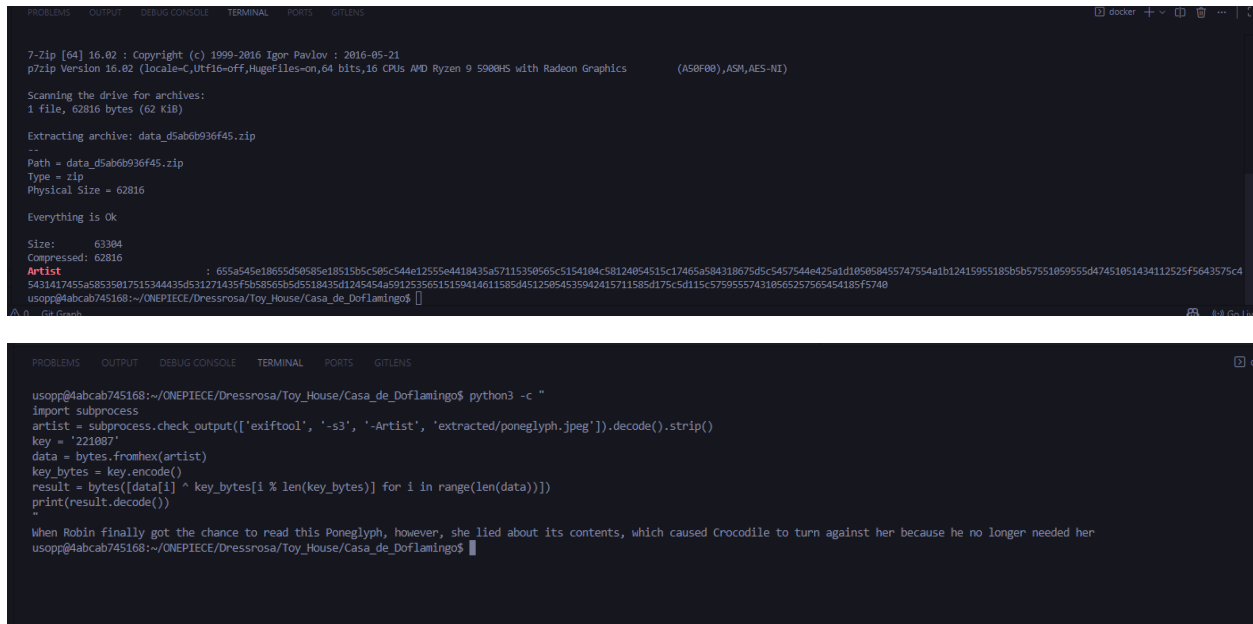
Comandos principales

```
cd ~/ONEPIECE/East_Blue/Syrup_Village/Casa_de_Merry
```

```
ls -la
```

```
cat flag.txt
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GIT LENS
usopp@4abcab745168:~/ONEPIECE/Dressrosa/Toy_House/Casa_de_Doflamingo$ cd ~/ONEPIECE/East_Blue/Syrup_Village/Casa_de_Merry
usopp@4abcab745168:~/ONEPIECE/East_Blue/Syrup_Village/Casa_de_Merry$ ls -la
total 16
drwxr-xr-x 1 usopp usopp 4096 May 24 03:51 .
drwxr-xr-x 1 usopp usopp 4096 May 24 03:51 ..
-rwxr-xr-x 1 usopp usopp 136 May 24 03:51 .marker.238
-rwxr-xr-x 1 usopp usopp 74 May 24 03:51 flag.txt
usopp@4abcab745168:~/ONEPIECE/East_Blue/Syrup_Village/Casa_de_Merry$ cat flag.txt
a50pp@4abcab745168:~/ONEPIECE/East_Blue/Syrup_Village/Casa_de_Merry$ python3 -c "abcab745168:~/ONEPIECE/East_Blue/Syrup_Village/Casa_de_Merry$ python3 -c "
import random
hex_string = open('flag.txt').read().strip()
cipher = bytes.fromhex(hex_string)
random.seed(1234)
keystream = bytes(random.randint(0, 255) for _ in range(len(cipher)))
plain = bytes(c ^ k for c, k in zip(cipher, keystream))
print(plain.decode())
"
FLAG_5bc1317c04f63305002507990ca9fab
usopp@4abcab745168:~/ONEPIECE/East_Blue/Syrup_Village/Casa_de_Merry$ cd ~/ONEPIECE/Dressrosa/Toy_House/Casa_de_Doflamingo
7z x -p"FLAG_aeb36ce81cda75a776a0088a4c8a35f2" data_d5ab6b936f45.zip -oextracted
0 / 0 - Git Graph
```



```
7-Zip [64] 16.02 : Copyright (c) 1999-2016 Igor Pavlov : 2016-05-21
p7zip Version 16.02 (locale=C,UTF16-off,HugeFiles-on,64 bits,16 CPUs AMD Ryzen 9 5900HS with Radeon Graphics (ASBF00),ASPI,AES-NI)

Scanning the drive for archives:
1 file, 62816 bytes (62 KiB)

Extracting archive: data_d5ab6936f45.zip
--
Path = data_d5ab6936f45.zip
Type = zip
Physical Size = 62816

Everything is Ok

Size:      61304
Compressed: 62816
Artist      : 65a545e1865d5d0585e18515b5c505c544e12555e4418435a57115350565c5154104c58124854515c17465a584318675d5c5457544e425a1d105058455747554a1b12415955185b5b57551059555d47451051434112525f5643575c4
5431417455a58535017515344435d31271435f5b58565b5d5518435d1245454a59125356515159414611585d45125054535942415711585d175c5d115c75955574318565257565454185f5748
usopp@4abcab745168:~/ONEPIECE/Dressrosa/Toy_House/Casa_de_DoFlamingo$

usopp@4abcab745168:~/ONEPIECE/Dressrosa/Toy_House/Casa_de_DoFlamingo$ python3 -c "
import subprocess
artist = subprocess.check_output(['exiftool', '-s3', '-Artist', 'extracted/poneglyph.jpeg']).decode().strip()
key = '221087'
data = bytes.fromhex(artist)
key_bytes = key.encode()
result = bytes([data[i] ^ key_bytes[i % len(key_bytes)] for i in range(len(data))])
print(result.decode())

When Robin finally got the chance to read this Poneglyph, however, she lied about its contents, which caused Crocodile to turn against her because he no longer needed her
usopp@4abcab745168:~/ONEPIECE/Dressrosa/Toy_House/Casa_de_DoFlamingo$
```

Resultados

Flag encontrada: **FLAG_5bc1317c04f633059025079960ca9fab**

Poneglyph extraído: ***"When Robin finally got the chance to read this Poneglyph, however, she lied about its contents, which caused Crocodile to turn against her because he no longer needed her"***

Análisis técnico — Reto Usopp (Stream Cipher custom)

Se navegó el laberinto hasta encontrar el flag.txt real por su contenido hexadecimal. El cifrado usa el generador pseudoaleatorio de Python con semilla fija 1234 para producir un keystream, que luego se combina con XOR contra el texto cifrado. Al conocer la semilla se puede reproducir exactamente el mismo keystream y revertir el cifrado. Para el poneglyph se extrajo el ZIP con la flag de Zoro como contraseña y se descifró el campo Artist con XOR usando el carné.

Reto 4 - Nami

ChaCha20

Objetivo: descifrar con ChaCha20. Clave 32 bytes y nonce 8 bytes derivados del carne.

Contraseña del ZIP: flag de Usopp.

Comandos principales

```
cd ~/ONEPIECE/Skypiea/Pumpkin_Cafe/Casa_de_Conis
```

```
ls -la
```

```
cd ~/ONEPIECE/East_Blue/Loguetown/Casa_de_Dragon
```

```
ls -la
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

nami@38ccd763c448:~$ cd ~/ONEPIECE/Skypiea/Pumpkin_Cafe/Casa_de_Conis
nami@38ccd763c448:~/ONEPIECE/Skypiea/Pumpkin_Cafe/Casa_de_Conis$ ls -la
total 64
drwxr-xr-x 1 nami nami 4096 May 24 03:51 .
drwxr-xr-x 1 nami nami 4096 May 24 03:51 ..
-rwxr-xr-x 1 nami nami 143 May 24 03:51 .marker_238
-rwxr-xr-x 1 nami nami 51924 May 24 03:51 data_652b63a0f61c.zip
nami@38ccd763c448:~/ONEPIECE/Skypiea/Pumpkin_Cafe/Casa_de_Conis$ cd ~/ONEPIECE/East_Blue/Loguetown/Casa_de_Dragon
nami@38ccd763c448:~/ONEPIECE/East_Blue/Loguetown/Casa_de_Dragon$ ls -la
total 16
drwxr-xr-x 1 nami nami 4096 May 24 03:51 .
drwxr-xr-x 1 nami nami 4096 May 24 03:51 ..
-rwxr-xr-x 1 nami nami 131 May 24 03:51 .marker_238
-rwxr-xr-x 1 nami nami 74 May 24 03:51 flag.txt
nami@38ccd763c448:~/ONEPIECE/East_Blue/Loguetown/Casa_de_Dragon$ cat flag.txt
ae5d2328f39fb6c41323ceec727f1a33911cbb720f1ecfc5ee0130718b217125a657c178nami@38ccd763c448:~/ONEPIECE/East_Blue/Loguetown/Casa_de_Dragon$
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

"

7-Zip [64] 16.02 : Copyright (c) 1999-2016 Igor Pavlov : 2016-05-21
p7zip Version 16.02 (locale=C,Utf16=off,HugeFiles=on,64 bits,16 CPUs AMD Ryzen 9 5900HS with Radeon Graphics (A50F00),ASM,AES-NI)

Scanning the drive for archives:
1 file, 51924 bytes (51 KiB)

Extracting archive: data_652b63a0f61c.zip
--
Path = data_652b63a0f61c.zip
Type = zip
Physical Size = 51924

Everything is Ok

Size:          52252
Compressed: 51924
the Tomb of the Kings where the Poneglyph was held became unstable and began collapsing around them.
nami@38ccd763c448:~/ONEPIECE/Skypiea/Pumpkin_Cafe/Casa_de_Conis$
```

Resultados

Flag encontrada: **FLAG_970a464429763b98a97c0e2893351c90**

Poneglyph extraido:

"the Tomb of the Kings where the Poneglyph was held became unstable and began collapsing around them."

"Análisis técnico: ChaCha20 usa operaciones ARX en 20 rondas sobre un estado de 512 bits, evitando ataques de cache-timing que afectan a AES. La seguridad del reto depende de una derivación de clave adecuada: repetir el clave es funcional pero no equivale a una KDF real como HKDF.

8. Resumen de resultados

Reto	Cifrado	Flag	Poneglyph
Luffy	XOR	FLAG_2c9ad67af4a916373725276ca6244b07	<i>"Crocodile targeted the Arabasta Kingdom because of its Poneglyph, which contained information on the whereabouts of Pluton,"</i>
Zoro	RC4	FLAG_aeb36ce81cda75a776a0088a4c8a35f2	<i>"and created the Baroque Works syndicate in an attempt to bring down the Arabasta Kingdom and claim Pluton, employing Robin to read the Poneglyph"</i>

Usopp	PRNG	FLAG_5bc1317c04f633059025079960ca9fab	<i>"When Robin finally got the chance to read this Poneglyph, however, she lied about its contents, which caused Crocodile to turn against her because he no longer needed her"</i>
Nami	ChaCha20	FLAG_970a464429763b98a97c0e2893351c90	<i>"the Tomb of the Kings where the Poneglyph was held became unstable and began collapsing around them."</i>

9. Reflexion final

Este proyecto fue una experiencia práctica que permitió entender cómo funcionan los cifrados simétricos más allá de la teoría. Al tener que navegar manualmente entre los archivos, identificar cuáles eran reales y cuáles falsos, y aplicar cada algoritmo de descifrado, se comprendió que la criptografía no solo se trata de fórmulas matemáticas, sino de saber cómo y dónde se aplica en un contexto real.

Algo importante que aprendí es que en un CTF no basta con buscar una bandera directamente, sino que hay que prestar atención a los detalles: el formato del contenido, los metadatos ocultos en imágenes, y la forma en que una flag depende de la anterior. Cada reto enseñó algo distinto, desde la simplicidad del XOR hasta la robustez de ChaCha20, dejando claro que la fortaleza de un cifrado depende tanto del algoritmo como de cómo se protege la clave.

10. Anexos

flags.txt:

Luffy: FLAG_2c9ad67af4a916373725276ca6244b07

Zoro: FLAG_aeb36ce81cda75a776a0088a4c8a35f2

Usopp: FLAG_5bc1317c04f633059025079960ca9fab

Nami: FLAG_970a464429763b98a97c0e2893351c90

poneglyphs.txt:

Luffy: Crocodile targeted the Arabasta Kingdom because of its Poneglyph, which contained information on the whereabouts of Pluton,

Zoro: and created the Baroque Works syndicate in an attempt to bring down the Arabasta Kingdom and claim Pluton, employing Robin to read the Poneglyph

Usopp: When Robin finally got the chance to read this Poneglyph, however, she lied about its contents, which caused Crocodile to turn against her because he no longer needed her

Nami: the Tomb of the Kings where the Poneglyph was held became unstable and began collapsing around